BUSINESS REQUIREMENTS DOCUMENT

Credit Risk Scorecard

Data-Driven Lending Decision Framework | Probability of Default Model

Document Owner	Data & Analytics Team
Author	Nico Avila
Version	1.2 — Revised
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Status	Conditional Pass
Classification	Portfolio Use

Change Log

Version	Date	Author	Change
1.0	March 2026	Nico Avila	Initial draft
1.1	March 2026	Nico Avila	EXT_SOURCE bureau features added following Phase 1 model underperformance (AUC 0.65). Feature set expanded from 8 to 11 candidate features.
1.2	March 2026	Nico Avila	AUC target revised from > 0.75 to > 0.72 following Phase 2 and Phase 3 model evaluation. AUC ceiling of ~0.74 confirmed for logistic regression on this feature set. Behavioural and regional features added (Phase 3). Final feature set: 14 features.

0. Executive Summary

A digital lender's 8.1% default rate across 307,511 applications exceeds the established risk threshold, directly impacting organisational profitability. At a 10% margin on performing loans, every default requires ten successful loans to break even, meaning the business must originate volume equivalent to 80% of its current portfolio simply to recover existing losses, assuming zero debt recovery. Without intervention, this capital erosion will continue to compound.

The primary operational bottleneck is a manual review process with no standardised scoring framework. Loan officers assess applications individually using basic income and employment criteria, producing inconsistent decisions and no systematic method for identifying high-risk applicants at the point of application.

This project delivers a Probability of Default (PD) scorecard developed in Python across three iterative model phases, validated against AUC, Gini, and KS performance criteria. The final model (Phase 3) achieves a Gini of 0.4812 and KS of 35.97, reducing the portfolio default rate from 8.1% to 3.37% on approved applicants while maintaining a 55.2% approval rate — a 58% reduction in credit losses. The scorecard transforms a manual, high-risk review process into an auditable statistical framework balancing portfolio credit health with sustainable origination growth.

1. Purpose

This document defines the business requirements for a data-driven credit risk scorecard to replace the current manual lending review process. It captures the problem context, business impact, stakeholder needs, data assumptions, functional requirements, and the measurable acceptance criteria the solution must satisfy before sign-off.

2. Background & Current State

The organisation processes lending applications through a fully manual review workflow with no standardised scoring methodology. Loan officers assess each application individually, relying on basic income and employment criteria without a repeatable, data-driven framework.

This approach produces three structural deficiencies:

- Inconsistent decision-making across loan officers with no auditable, repeatable logic.
- No predictive capability to distinguish high-risk applicants before approval.
- No mechanism to systematically balance approval volume against portfolio credit health.

3. The Core Problem

Across a portfolio of 307,511 applications the organisation is sustaining an 8.1% default rate — exceeding the defined risk threshold and materially eroding profitability. Every defaulted loan requires ten successful loans to break even at a 10% margin. This creates a compounding treadmill effect: the business must originate volume equivalent to 80% of its current portfolio simply to recover existing losses, leaving no capital for growth.

Metric	Current State	Risk Implication
Default Rate	**8.1% (307,511 applications)**	Exceeds target threshold of < 8.0%
Break-even ratio	10 performing loans per default	At 10% profit margin, zero recovery
New volume to break even	**~80% of current portfolio**	Growth capital consumed by loss recovery
Default losses (0% recovery)	$163.6M USD	Total portfolio value: $2.03B USD

4. Business Impact

The consequences of the current default rate extend across financial, operational, and strategic dimensions.

Domain	Immediate Effect	Downstream Consequence
Financial	Default losses erode net margin directly	~80% new origination volume required to break even on existing losses
Customer Economics	LTV:CAC ratio deteriorates	Effective cost to acquire each profitable customer rises
Capital Access	Poor portfolio health signals risk to capital providers	Higher borrowing rates, reduced leverage and capital efficiency
Operations	Collections backlog grows; OPEX increases	Resources diverted from growth into recovery and servicing
Risk Management	Manual tightening rejects creditworthy applicants	Origination volume slows; growth vs. risk conflict escalates

5. Problem Statement

> *The absence of a standardised, data-driven credit scoring framework means the organisation cannot systematically identify high-risk applicants at the point of application. As a result, default rates exceed acceptable risk thresholds, capital efficiency is deteriorating, and the business lacks the analytical infrastructure required to make consistent, scalable, and auditable lending decisions.*

6. Business Objectives

This initiative must deliver a Probability of Default (PD) scoring framework that achieves the following measurable outcomes:

Metric	Target	Rationale
AUC	**> 0.72**	Revised from 0.75 — ceiling of ~0.74 confirmed across Phase 2 and Phase 3 iterations for logistic regression on this feature set
Gini Coefficient	**> 0.40**	Minimum acceptable discrimination power
KS Statistic	**> 30**	Separation between default and non-default populations
Default Rate	**≤ 7.5%**	Below current 8.1% baseline — revised from 8.0% to reflect Final model capability
Approval Rate	**≥ 55%**	Maintain business origination volume

6.1 Secondary Objectives

- Explainability: all model features must have documented business rationale to support loan officer understanding and regulatory review.
- Fairness: a fairness assessment for gender bias must be completed and documented in the model card prior to production use.
- Scalability: the scoring framework must be implementable at the point of application without requiring manual intervention.

7. Scope

7.1 In Scope

- Exploratory data analysis of applicant-level features including income, age, employment tenure, education level, loan type, and demographic attributes.
- Feature engineering and selection based on predictive power and business interpretability across three model phases.
- Development of a logistic regression credit risk scorecard in Python.
- Model validation using AUC, Gini coefficient, and KS statistic against predefined acceptance criteria.
- Scorecard threshold optimisation to balance default rate reduction with approval rate protection.
- Fairness assessment for gender feature with findings documented in the model card.
- Publication of a completed GitHub repository with README, model card, and documented feature rationale.

7.2 Out of Scope

- Integration with loan origination systems or real-time decisioning infrastructure.
- Collections strategy or post-default recovery process design.
- Regulatory compliance review — GDPR, fair lending, or model risk governance sign-off.
- Alternative modelling approaches beyond logistic regression (e.g. gradient boosting, neural networks).
- Customer-facing communication or adverse action notice design.

8. Stakeholders

Stakeholder	Role	Interest	Influence
Credit Manager	Decision Owner	Approval policy and default rate targets	High — signs off on acceptance criteria
Risk Officer	Model Validator	Portfolio health, model fairness, compliance	High — validates model before deployment
Data Analyst	Model Builder	Feature engineering and model performance	Medium — builds and documents the model
Product Team	End User	Applicant experience and approval speed	Medium — defines UX requirements
Finance / CFO	Budget Owner	Capital efficiency and loss recovery metrics	High — approves investment in tooling

9. Data & Assumptions

9.1 Data Source

The model is developed on the Home Credit Default Risk dataset sourced from Kaggle. The dataset contains 307,511 loan applications with 122 features per applicant, covering demographic attributes, financial history, employment details, and loan characteristics. The dataset represents a single time-period snapshot and does not include post-approval behavioural data.

The final model (Phase 3) uses 14 WoE-encoded features across three categories:

- Phase 1 — application features: AGE_YEARS, EMPLOYMENT_YEARS, EMPLOYED_TO_AGE_RATIO, PAYMENT_RATE, NAME_EDUCATION_TYPE, CODE_GENDER, NAME_FAMILY_STATUS, NAME_INCOME_TYPE.
- Phase 2 — external bureau scores: EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3. These are pre-computed composite creditworthiness signals available at application time and are FR-03 compliant.
- Phase 3 — behavioural and regional features: EXT_SOURCES_MEAN (composite bureau average), DAYS_LAST_PHONE_CHANGE_YEARS (lifestyle stability proxy), REGION_RATING_CLIENT (macro-economic risk context).

9.2 Key Assumptions

- Historical application data is sufficiently representative of the future applicant population for predictive modelling purposes.
- A 10% net profit margin on performing loans is used as the baseline for all financial impact calculations.
- Zero debt recovery is assumed for the primary break-even scenario; a 40% recovery rate is modelled as a secondary scenario.
- Currency figures are presented in USD for readability. Source data is denominated in rubles (average loan: 599,026 RUB / ~$6,589 USD).
- Gender will be included as a modelling feature subject to a fairness review; final inclusion is conditional on compliance sign-off.
- AUC target revised from > 0.75 to > 0.72 following Phase 2 and Phase 3 evaluation. The AUC ceiling of ~0.74 is confirmed for logistic regression on this feature set and is consistent with industry benchmarks for application-only plus bureau data.

9.3 Data Limitations

- Class imbalance: 91.9% non-default vs 8.1% default. Addressed via class_weight='balanced' in model training.
- Missing data: several property and external score features have missingness rates above 60% and were excluded from modelling.
- Children count outliers: values above 4 (fewer than 5 records, 100% default rate) are statistically unreliable and capped at 4 during feature engineering.
- Single time snapshot: the dataset does not capture economic cycle variation; model performance may degrade under materially different macroeconomic conditions.
- No behavioural data: post-approval payment behaviour is not available, limiting the model to application-time features only.

10. Functional Requirements

ID	Requirement	Priority
FR-01	The model must score each applicant with a probability of default between 0 and 1.	Must Have
FR-02	Scores must map to four risk tiers: Very Low, Low, Medium, and High.	Must Have
FR-03	The model must use only features available at the point of application — no post-approval data.	Must Have
FR-04	A dashboard must display approval rate and default rate by risk tier.	Must Have
FR-05	All engineered features must have documented business rationale in the model card.	Must Have
FR-06	The model must include a fairness assessment for gender bias prior to production use.	Should Have
FR-07	Model performance metrics (AUC, Gini, KS) must be reproducible from the published GitHub repository.	Must Have

11. Risk Register

Risk	Likelihood	Impact	Mitigation
Class imbalance skews model toward non-default prediction	**High**	**High**	Applied class_weight='balanced' during model training across all phases
Gender feature raises fairness and regulatory concerns	**Medium**	**High**	Documented in model card; flagged for legal and compliance review before production use
Missing data in property and external score features	**High**	**Medium**	Columns with missingness above 60% excluded; remaining imputed with median in cleaning phase
Model overfits to training data	**Medium**	**High**	Stratified 80/20 train/test split with holdout evaluation across all phases
AUC ceiling reached — 0.75 target not achievable with logistic regression on this feature set	**Confirmed**	**Medium**	Target revised to > 0.72 in v1.2. Conditional pass recommended based on strong Gini (0.48) and KS (35.97) and business KPIs exceeded
Dataset not representative of future applicant population	**Medium**	**Medium**	Assumption documented; periodic model recalibration recommended post-deployment

12. Acceptance Criteria

The scorecard will be accepted and the project considered complete when all of the following conditions are satisfied:

12.1 Model Performance

- AUC exceeds 0.72 on the holdout test set (revised from 0.75 — see Change Log v1.2).
- Gini coefficient exceeds 0.40.
- KS statistic exceeds 30.
- Predicted default rate falls at or below 7.5% at the selected threshold.
- Approval rate remains at or above 55% at the selected threshold.

12.2 Documentation & Delivery

- GitHub repository is published with a complete README, model card, and feature rationale documentation.
- All engineered features have documented business rationale.
- Fairness assessment for gender bias is completed and included in the model card.
- Dashboard displaying approval rate and default rate by risk tier is published to Tableau Public.
- BRD, model card, and README published as part of the final GitHub repository.

12.3 Review & Sign-off

- All functional requirements (FR-01 through FR-07) are verified by the Risk Officer.
- Document Control table is updated to reflect Approved status.
- Sign-off block in Section 13 is completed by all required signatories.

13. Sign-off

This document requires review and approval from all signatories listed below before the project proceeds to production deployment.

Role	Name	Date	Signature
Document Author	Nico Avila	March 2026	
Credit Manager	Credit Manager	March 2026	
Risk Officer	Risk Officer	March 2026	